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Article

# Tension and Force-Resistant Attachment Are Essential for Myofibrillogenesis in *Drosophila* Flight Muscle

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## Summary

**Background:** Higher animals generate an elaborate muscle-tendon network to perform their movements. To build a functional network, developing muscles must establish stable connections with tendons and assemble their contractile apparatuses. Current myofibril assembly models do not consider the impact of muscle-tendon attachment on myofibrillogenesis. However, if attachment and myofibrillogenesis are not properly coordinated, premature muscle contractions can destroy an unstable myotendinous system, leading to severe myopathies.
**Results:** Here, we use *Drosophila* indirect flight muscles to investigate how muscle-tendon attachment and myofibrillogenesis are coordinated. We find that flight muscles first stably attach to tendons and then assemble their myofibrils. Interestingly, this myofibril assembly is triggered simultaneously throughout the entire muscle, suggesting a self-assembly mechanism. By applying laser-cutting experiments, we show that muscle attachment coincides with an increase in mechanical tension before periodic myofibrils can be detected. We manipulated tension buildup within the myotendinous system either by genetically compromising attachment initiation and integrin recruitment to the myotendinous junction or by optically severing tendons from muscle. Both treatments cause strong myofibrillogenesis defects. We find that myosin motor activity is required for both tension formation and myofibril assembly, suggesting that myofibril assembly itself contributes to tension buildup.
**Conclusions:** Our results demonstrate that force-resistant attachment enables a stark tension increase in the myotendinous system. Subsequently, this tension increase triggers simultaneous myofibril self-assembly throughout the entire muscle fiber. As myofibril and sarcomeric architecture as well as their molecular components are evolutionarily conserved, we propose a similar tension-based mechanism to regulate myofibrillogenesis in vertebrates.

## Introduction

Higher animals utilize muscles to move their skeleton and thus their entire body during daily life. The force to power these movements is produced within the muscles and transmitted via tendons to the skeleton. To build a functional unit, muscle, tendon, and skeleton development need to be precisely coordinated. In both insects and vertebrates, myoblasts fuse to myotubes that migrate toward their tendon cell partners. Next, myotubes attach to tendons in order to establish a stable connection, which often lasts for the rest of the animal's life (reviewed in [1, 2]). After successful tendon attachment, the myotube is gradually converted into a myofiber by a process called myofibrillogenesis, during which each myofiber assembles many, often hundreds of, myofibrils. Each individual myofibril spans from one attached muscle end to the other and contains a large number of stereotyped contractile units, the sarcomeres, with alternating Z discs and M lines. After myofibrillogenesis has been completed, the muscle fiber is ready to contract and can move the skeleton by pulling on stably connected tendons [3, 4].

How are myofibrils assembled de novo? Theoretical models have postulated an essential role for myosin-based contractility in assembling actin filaments to a sarcomere-like pattern [5]. This assembly is thought to generate mechanical tension within the developing muscle. As muscle attachment to tendons is force resistant, it is tempting to hypothesize that muscle-tendon attachment is a prerequisite for tension buildup, which may in turn trigger myofibrillogenesis. However, to date we have an incomplete mechanistic understanding of how myotubes attach to tendons and whether or not this attachment generates mechanical tension within the myotendinous system.

A few years ago, we identified Kon-tiki (Kon), a single-pass transmembrane molecule of the neurexin family, as primarily expressed and required to initiate muscle-tendon attachment in one class of *Drosophila* embryonic muscles, the ventral-longitudinal muscles [6]. *kon* genetically interacts with the extracellular matrix (ECM) component laminin B2 [7] and αPS1-integrin (*mew*) present on tendon cells [8]. However, it is unclear whether Kon binds to either of these molecules and whether initiation of attachment by Kon is a general mechanism required in different muscle types.

*Drosophila* as well as vertebrate studies have established that integrins are the main surface molecules mediating force-resistant attachment of muscle and tendons to ECM, which is a main component of mature myotendinous junctions. Consequently, integrins connect mature muscles via ECM to tendons (reviewed in [9, 10]). Integrins are heterodimers consisting of one α and one β subunit. In *Drosophila*, loss of either βPS-integrin (β-*int*, *mys*) or both α subunits, αPS1-integrin and αPS2-integrin, leads to a muscle-tendon detachment phenotype in all body muscles [11, 12], demonstrating the need to establish force-resistant muscle-tendon attachment prior to muscle contraction.

The exact process of myofibrillogenesis is controversial (reviewed in [13–15]). The classical premyofibril model proposes that short, irregular premyofibrils mature to regular myofibrils by the gradual exchange of nonmuscle myosin II with muscle myosin II [16]. Because the first myofibrils are often found close to the cell membrane [17] and integrins have been suggested to play an important role in myofibrillogenesis in fly embryos and mice [18–20], it has been speculated that integrin adhesion sites along the myofiber surface may promote assembly of premyofibrils, which then spread

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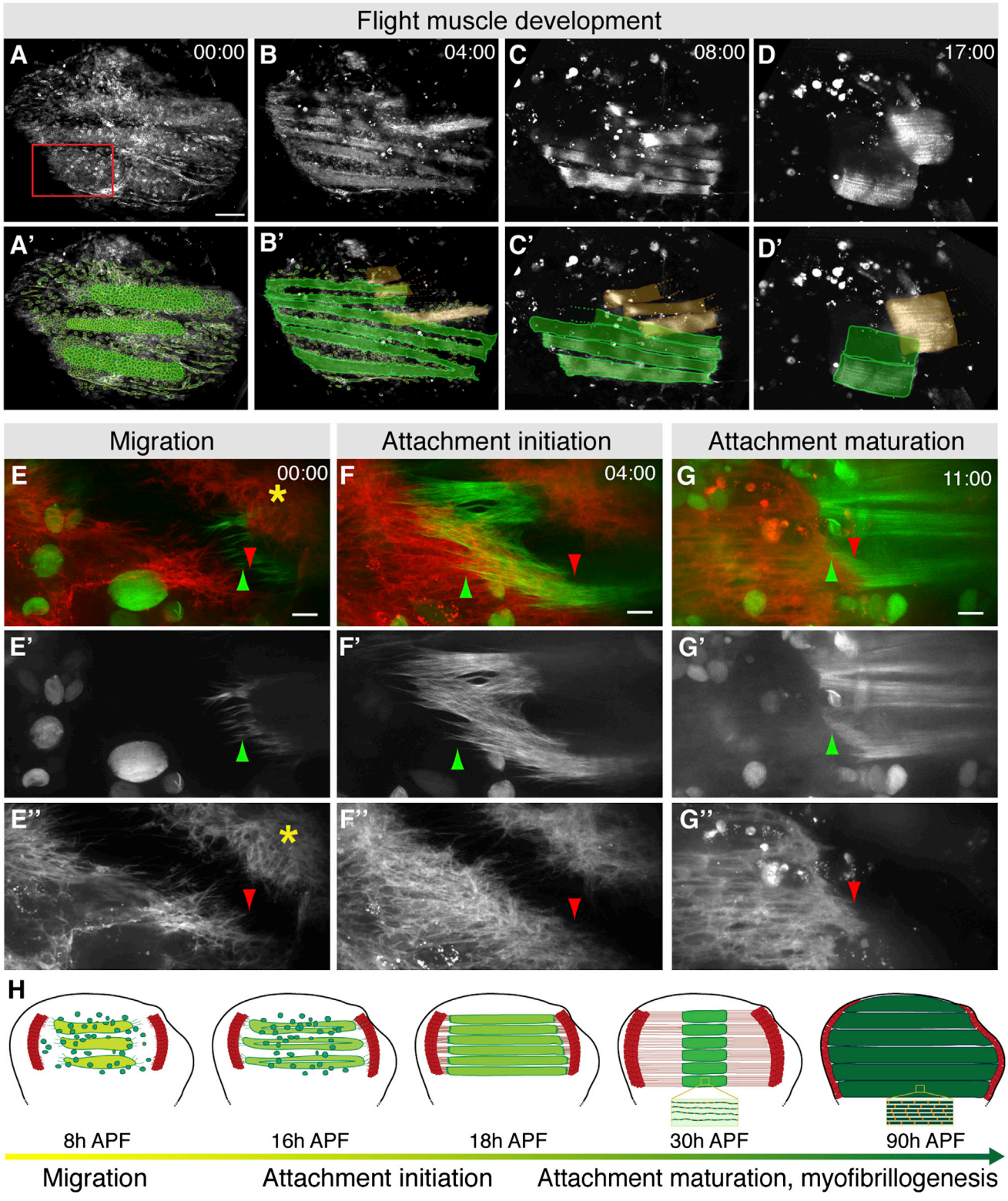



Figure 1. Indirect Flight Muscle Development: Migration and Attachment

(A–D) Time points from a multiphoton movie (Movie S1) using *1151-GAL4,UAS-CD8-GFP* to label myoblasts, dorsal-longitudinal flight muscles (DLMs), and dorsoventral flight muscles (DVMs). Myoblasts and DLMs are colored in green and DVMs in brown in (A′)–(D′). DLMs move ventrally over time, and only the dorsal ones remain visible in (C) and (D). The red box indicates the magnification shown in Figures S1A–S1D.

*(legend continued on next page)*

throughout the entire muscle fiber [21]. Alternative models suggest that thin and thick filaments assemble independently from each other before subsequently interdigitating to form a functional sarcomere [22, 23]. Even multiple preformed complexes have been suggested to assemble to sarcomere-containing myofibrils [24]. Theoretical models of sarcomere formation predict the coordinated, simultaneous assembly of unordered actin and myosin filaments into periodic structures by actin plus-end filament crosslinking and myosin filament-dependent force generation, leading to rapid sorting of the antiparallel actin filaments [5, 25, 26]. Combining these theoretical and experimental assembly models with the suggested role of integrins as nucleators of premyofibrils and the discovery that integrin adhesion sites can be under mechanical tension [27], it is enticing to speculate that mechanical tension within the myotendinous system could contribute to myofibril assembly in vivo.

*Drosophila* indirect flight muscles (IFMs) have a particularly regular myofibrillar architecture that allows fast, asynchronous, stretch-activated contraction cycles powering flight [28, 29]. IFMs consist of two antagonistic muscle sets: the dorsal-longitudinal flight muscles (DLMs), running anterior to posterior through the thorax, and the dorsoventral flight muscles (DVMs), oriented from dorsal to ventral. During pupal development, DVMs form de novo by myoblast fusion, whereas DLMs use remodeled larval muscles as templates to which myoblasts fuse [30, 31]. In both cases, the resulting myotubes attach to *stripe*-expressing tendon cells located within the developing adult epidermis, leading to a stably connected muscle-tendon network [32]. Electron microscopic studies of attached myofibers at different time points have shown the abrupt appearance of long myofibrils without prior detection of premyofibrils [33, 34], thus challenging the premyofibril model.

In this study, we set out to investigate the processes that enable attachment of muscles to tendons and test whether this attachment results in the buildup of tension in order to trigger myofibrillogenesis. We chose *Drosophila* IFMs as they are particularly well suited for live imaging of attachment and myofibrillogenesis, as well as for the application of laser-induced microlesions to quantify and manipulate the formation of mechanical tension within muscle-tendon tissue. We demonstrate that mechanical tension is generated during attachment and plays a key role in myofibrillogenesis.

## Results

### Three Phases of Muscle-Tendon Migration and Attachment

Attachment of muscles to their tendons has not been carefully investigated in flight muscles. Therefore, we first characterized the process of muscle-tendon attachment and subsequently tested whether this attachment allows for the buildup of tension. We performed live multiphoton imaging of intact pupa expressing globular moesin actin-binding domain tagged with GFP (GFP-Gma) in developing flight muscles and focused our analysis on DLMs, as they are closest to the pupal surface. After head eversion at about 8–10 hr after puparium formation (APF), myoblasts migrate to templates of the DLMs and begin to fuse with them (Figure 1A; see also Movie S1 available online). Myoblast fusion triggers splitting of the three DLM templates, resulting in six myotubes that start to compact along their long axis to about half their original length, before they eventually elongate again (Figures 1B–1D, 1H; Movie S1). During myoblast fusion and DLM splitting, dynamic filopodial extensions longer than 20 μm form at the myotube tips and preferentially point toward the basal side of the tendon epithelium. (Figures S1A–S1C; Movie S2). After myotube splitting, these filopodia gradually disappear and the myotube surface becomes smooth (Figure S1D). Colabeling of tendon cells showed that tendons form long cellular extensions reaching up to 200 μm when myotubes are maximally compacted (Figures S1E–S1J; Movie S3). Taken together, these data show that myotubes change their behavior dramatically during attachment initiation. Their tips convert from highly dynamic exploratory behavior to a stable and well-attached state.

To directly observe attachment initiation at high resolution, we performed two-color spinning-disk confocal microscopy using *UAS-palmitylated-Cherry;stripe-GAL4* to label the tendon cell membrane and *mhc-TAU-GFP* to mark the forming DLMs (Movie S4). We found that DLMs extend their tips directly toward the basal side of their tendon cells, ignoring a closely located tendon group to which DVMs will attach (Figure 1E). Both myotube and tendon tips dynamically interdigitate for more than 5 hr, during which DLMs split and initiate attachment (Figure 1F; Movie S4). During the next 5 hr, filopodial dynamics at the myotendinous junction decrease, indicative of the maturation of myotube-tendon attachment. Subsequently, stably attached myotubes compact and tendon extensions elongate (Figure 1G; Movie S4).

Combining these in vivo imaging data, we define three distinct phases of myotube-tendon morphogenesis:

(1) Migration: myotubes form dynamic filopodial extensions at their tips and migrate directly toward their specific tendon targets.
(2) Attachment initiation: myotube and tendon tips interdigitate extensively, recognize each other, and initiate attachment.
(3) Attachment maturation: muscle tips become smooth and stably connected to tendons. Muscles compact in length, while tendons form long cellular extensions.

### Mechanical Tension Is Generated during Attachment

The dramatic elongation of tendon cells during muscle compaction suggests production of mechanical tension within the myotendinous system. To test this hypothesis, we characterized mechanical tension within the myotendinous system over a time period of 10 hr. We severed tendon extensions using a pulsed UV laser by applying a 2 μm cut orthogonal to the muscle-tendon axis and measured tissue recoil. We used pupae expressing membrane-tethered GFP in muscles and tendons at 13 hr, 18 hr, and 22 hr APF, representing stages of myotube migration, attachment initiation, and maturation,

(E–G) Time points taken from a *mhc-TAU-GFP;UAS-palm-Cherry;stripe-GAL4* movie (Movie S4) during migration (E), attachment initiation (F), and attachment maturation (G). Myotube ends are marked with a green arrowhead and tendon ends with a red arrowhead. A tendon field to which DVMs will attach is marked with an asterisk.
(H) Scheme of DLM development. Tendons are shown in red, myoblasts and DLMs in green. Zoom depicts myofibrillar organization at 30 hr and 90 hr after puparium formation (APF).
Time is indicated in hr:min. Scale bars represent 50 μm in (A)–(D) and 10 μm in (E)–(G).

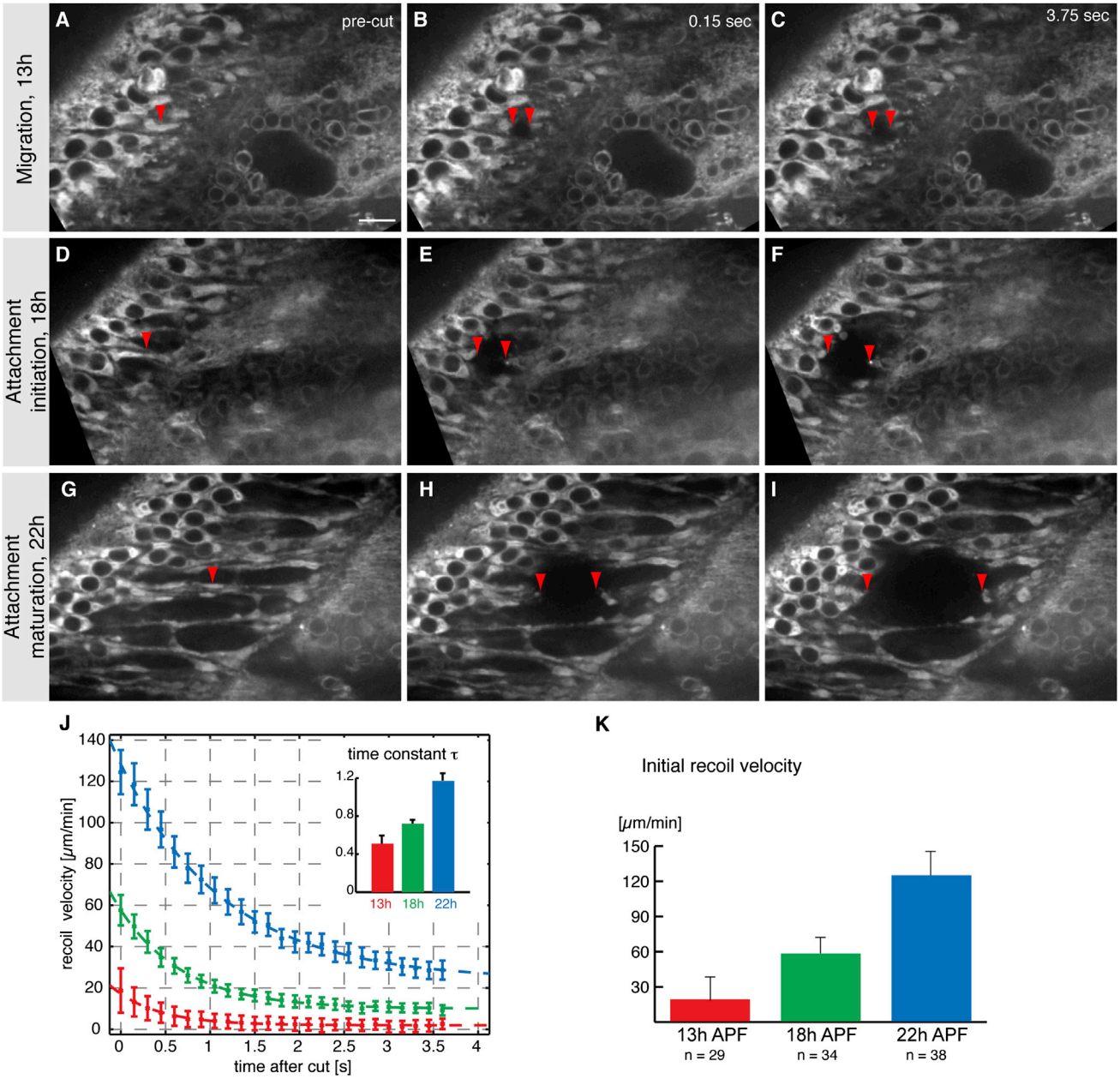



Figure 2. Mechanical Tension Increases during Muscle Attachment

(A–I) Time points from time-lapse movies of *UAS-CD8-GFP;Mef2-GAL4,stripe-GAL4*-expressing pupae subjected to laser cuts in tendon tissue at 13 hr APF migration stage (A–C), 18 hr APF attachment initiation (D–F), and 22 hr APF attachment maturation (G–I). Cut position and recoiling tendon fragments are labeled with arrowheads. Scale bar represents 10 μm.

(J) Orthogonal recoil velocities as a function of time, depicted as individual values (including error bars), and the decay curve fit to calculate the time constant τ of the decay of recoil velocity (within a 95% confidence interval).

(K) Initial recoil velocity at 13 hr, 18 hr, and 22 hr APF. Error bars represent SEM.

respectively. After cutting, tendon tissue adjacent to the cut moves in an orthogonal direction away from the cutting line (Figures 2A–2I; Movie S5). We calculated the recoil velocity of the tendon movement after cutting using fast imaging combined with automatic tracking and found an exponential decay over time, typical for a simple viscoelastic response (Figure 2J). Using the first and second frame after cutting, we measured the initial recoil velocity, which depends on both the tension present in the myotendinous system and the friction-like resistance that the elongations experience with the surrounding material during recoil. At the end of myotube migration at 13 hr APF, the initial recoil velocity is very low (18.7 ± 10.7 μm/min). During attachment initiation, however, it increases threefold (57.6 ± 7.3 μm/min) and doubles again during attachment maturation (124.5 ± 10.7 μm/min; Figure 2K). This increase in recoil velocity could be due to either an increase in mechanical tension during attachment maturation or a decrease in friction between elongations and extracellular material. Concomitantly with the increase in recoil velocity, we measured an increase of the time constant τ of

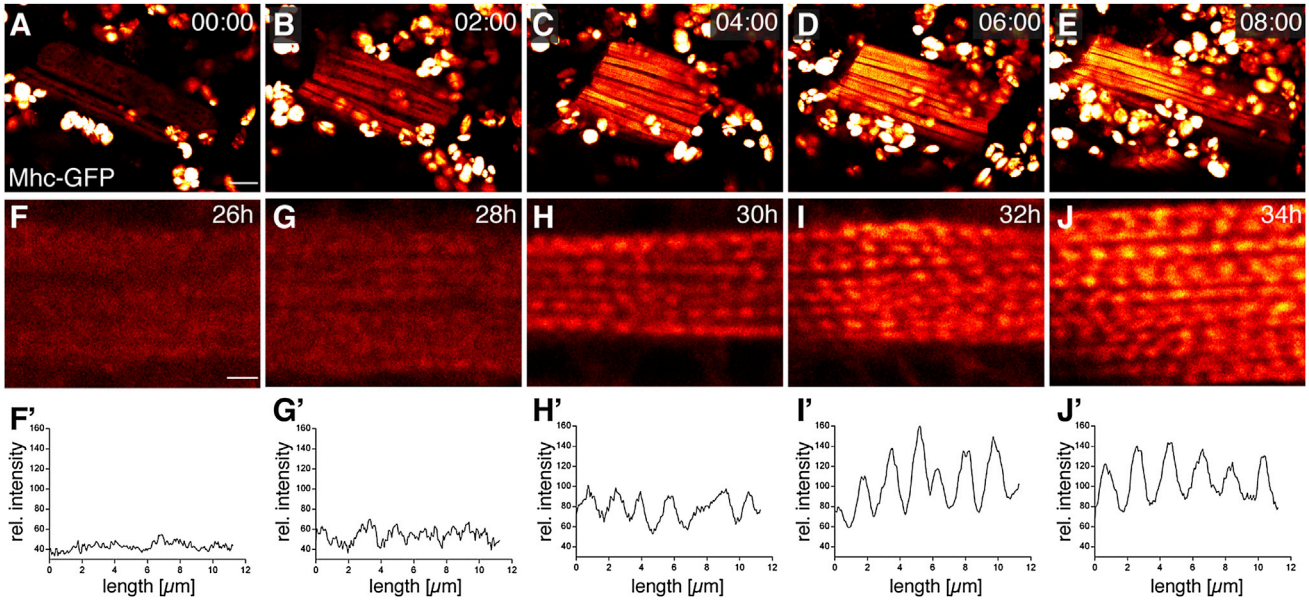



Figure 3. Simultaneous Myofibrillogenesis in Indirect Flight Muscles

(A–E) Time points from a multiphoton movie showing developing DLMs in pupae labeled with *Mhc-GFP* (*Mhc*$^{Wee\text{-}P26}$) beginning at 26 hr APF; time is indicated in hr:min. Note the simultaneous assembly of a periodic Mhc pattern.

(F–J) High-resolution single time points of DLMs from *Mhc-GFP* pupae at the indicated times.

(F′–J′) Relative intensity plots of a single representative developing myofibril at the respective time points in (F–J).

Scale bars represent 25 μm in (A)–(E) and 2 μm in (F)–(J).

the decay of recoil velocity over time (Figure 2J). This indicates a change in the material properties of the tendon extensions and their surrounding material during attachment maturation and is consistent with either an increase in friction between elongations and extracellular material or a softening of the extensions (i.e., decrease of extension elasticity). Both an increase in friction and a softening of the tissue during attachment maturation together with the measured increase in recoil speed are consistent with a strong increase in tension during attachment maturation. This demonstrates that the myotendinous system is indeed generating mechanical tension during attachment initiation that strongly increases during attachment maturation.

### Myofibrillogenesis Is Simultaneous throughout the Muscle Fiber

Based on in vitro and modeling studies, it was hypothesized that tension may influence myofibrillogenesis [5, 35–37]. Since we detected a strong increase in tension during attachment, we investigated whether this tension increase correlates with myofibrillogenesis. To analyze myofibrillogenesis in DLMs, we imaged endogenous GFP-tagged muscle myosin heavy chain (Mhc) starting at 26 hr APF using live multiphoton microscopy. At 26 hr APF, low levels of Mhc are homogenously distributed in the cytoplasm of the myofiber (Figures 3A and 3F; Movie S6). During myofiber compaction, this homogenous distribution is dramatically reorganized into a readily visible, periodically dotted Mhc array at 30 hr APF (Figures 3C and 3H; Movie S6). This array is further refined into distinct myosin filaments spaced at regular distances along a forming myofibril until 34 hr APF (Figures 3E and 3J; Movie S6). Strikingly, this periodic array appears simultaneously throughout the entire myofiber, which is about 150 μm long (Figures 3C–3E and 3H–3J). Mhc-containing aggregates are detectable in the center of the fiber at the same time as at the myofiber ends or surface. This indicates that Mhc is recruited or assembled into myofibrils without former localization to distinct short premyofibrils, which were proposed to originate from the fiber surface [17]. However, it is possible that fibril assembly occurs at or close to plasma membrane, which could be involuted to central parts of the fiber. The surprising speed with which the Mhc fibrillar array is formed and its periodic regularity suggest a mechanism of self-organization, consistent with theoretical models of myofibril assembly [26].

### *kon-tiki* Is Essential for Flight Muscle-Tendon Attachment

To test whether mechanical tension influences myofibrillogenesis, we interfered with tension formation by genetically manipulating muscle-tendon attachment. To identify genes that regulate flight muscle attachment, we investigated 186 candidate genes with a predicted transmembrane domain previously identified as pupal lethal in a muscle-specific RNAi screen [38]. Interestingly, muscle-specific knockdown of *kon* using *Mef2-GAL4* results in no or very few rounded IFMs at 90 hr APF (Figures 4A–4C). We confirmed this phenotype with two different hairpin constructs targeting distinct regions of the *kon* gene (*UAS-kon-IR-NIG* and *UAS-kon-IR-GD*), demonstrating the specificity of the phenotype. Additionally, we tested a third hairpin (*UAS-kon-IR-KK*) that resulted in a weaker phenotype with thinner DLMs compared to wild-type (Figure 4D). Our results show that *kon* is required for proper IFM morphogenesis with a possible role in attachment, since the remaining IFMs display a rounded morphology.

To identify the stage at which *kon* is required, we performed multiphoton live imaging of *kon* knockdown IFMs. We found that both myoblast fusion and myotube migration were normal (Figures S4A, S4F, and S4K); however, the first defects appeared during attachment initiation (Movie S7). Compared

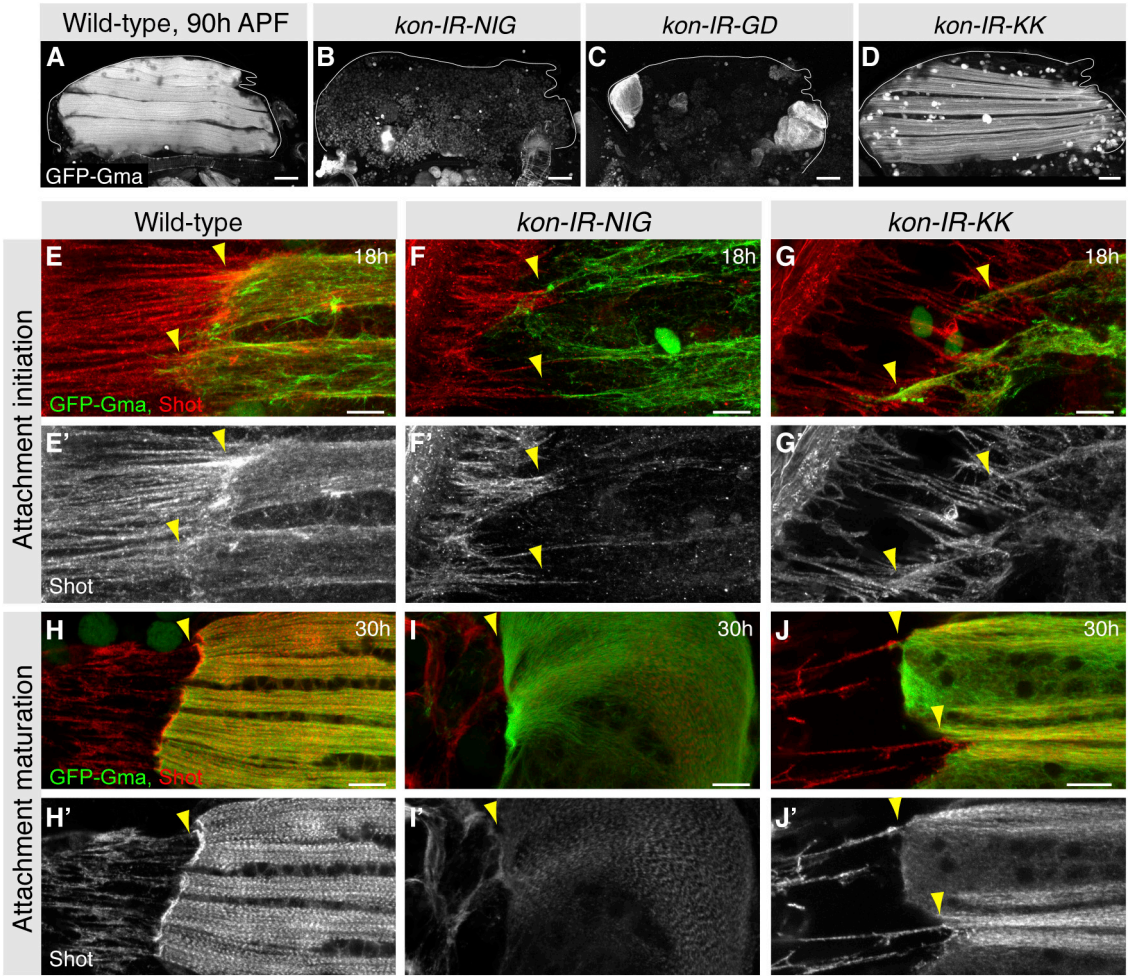



Figure 4. *kon* Is Essential to Initiate Myotube-Tendon Attachment

(A–D) Hemithorax of 90 hr APF pupae expressing *Mef2-GAL4,UAS-GFP-Gma* to label DLMs in wild-type (A) or in *UAS-kon-IR-NIG* (B), *UAS-kon-IR-GD* (C), or *UAS-kon-IR-KK* (D) knockdown pupae.

(E–J) Muscle-tendon morphology. Wild-type (E and H), *UAS-kon-IR-NIG* (F and I), and *UAS-kon-IR-KK* pupae (G and J) expressing *Mef2-GAL4,UAS-GFP-Gma* at 18 hr APF attachment initiation (E–G) and 30 hr APF maturation (H–J) are shown. DLMs are labeled with GFP (green) and tendons are labeled with anti-Shot (red); Shot also labels muscles. Arrowheads point to myotube-tendon contacts.

Scale bars represent 50 μm in (A)–(D) and 10 μm in (E)–(J).

to wild-type, *kon* knockdown myotube tips formed extremely long extensions (up to 70 μm) toward the tendon cells (Figures S2B, S2G, S2L, and S2M) but appeared unable to initiate attachment. During the stage of myotube compaction in wild-type, *kon* knockdown myotubes either rounded up, leading to unattached, rounded muscles (Figures S2C–S2E and S2H–S2J), or displayed rounded myotube ends and abnormally long myofibers in the *kon* hypomorph (Figures S2N and S2O–S2R). Therefore, we conclude that attachment initiation of DLMs to tendon cells is defective upon *kon* knockdown in muscle.

To investigate the myotube-tendon interface in detail, we stained tendons with the spectraplakin homolog Shot [39, 40] and labeled myotubes with *GFP-Gma*. Consistent with our time-lapse data, we found extensive interaction in wild-type between myotube and tendon tips during attachment initiation (18 hr APF), with prominent Shot accumulation at the myotendinous junction (Figure 4E). However, *kon* knockdown pupae showed only rare myotube-tendon contacts and no Shot accumulation at the junction (Figure 4F). *kon* knockdown muscles formed filopodial extensions directed toward their tendons but appeared unable to recognize them, and thus attachment was not initiated. As a consequence, *kon* knockdown myotubes completely rounded up during the compaction phase, sometimes with one end remaining close to the tendons (Figures 4H and 4I). The hypomorphic *kon* knockdown muscles also displayed reduced tendon contacts at 18 hr APF (Figure 4G), resulting in only patchy muscle-tendon attachment at 30 hr APF (Figure 4J). Taken together, these data suggest that *kon* is important for initiation of stable force-resistant attachment of IFMs to their respective tendons.

## Kon Is Required for Integrin Accumulation at the Myotendinous Junction

Kon and integrins are both located at the myotendinous junction of embryonic muscles [6], at which integrins facilitate force-resistant attachment by recruiting the adaptor Talin to the tail of βPS-integrin [41]. To dissect the molecular interplay of Kon and integrin during attachment, we investigated

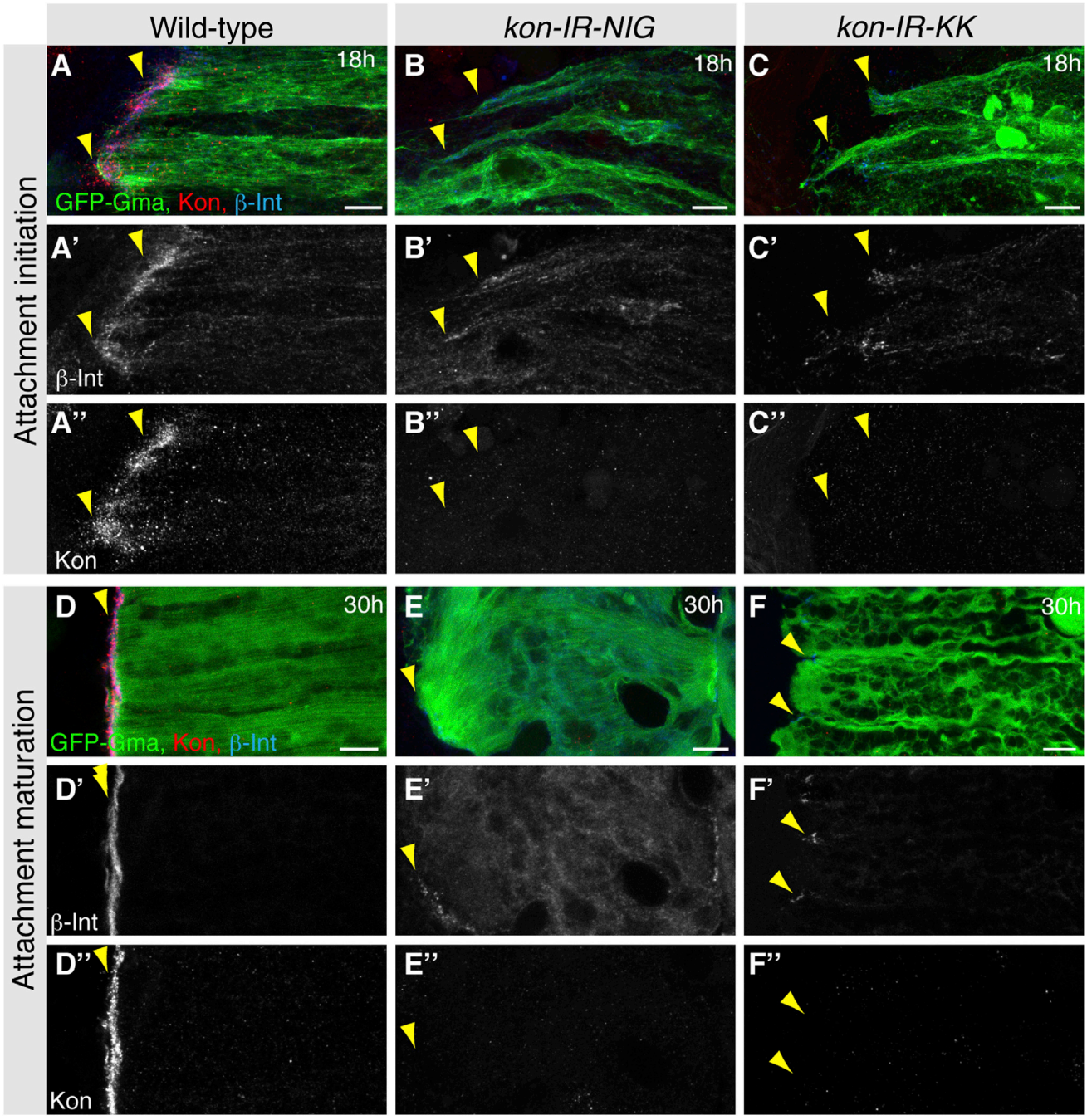



Figure 5. *kon*-Dependent Attachment Initiation Is Essential for Integrin Localization

Dissected wild-type (A and D), *UAS-kon-IR-NIG* (B and E), and *UAS-kon-IR-KK* pupae (C and F) expressing *Mef2-GAL4,UAS-GFP-Gma* at 18 hr APF attachment initiation (A–C) and 30 hr APF maturation (D–F), stained for GFP to label DLMs (green), anti-Kon (red), and anti-βPS-Int (anti-βPS, blue). Note the prominent localization of Kon and integrin at myotube tips during attachment initiation and maturation, both of which are largely lost upon *kon* knockdown (arrowheads). Scale bars represent 10 μm.

the localization of Kon, βPS-integrin (β-Int, *mys*), Talin, Thrombospondin (Tsp, an essential integrin ligand), and phosphorylated FAK (pFAK, a readout of integrin signaling) [42]. We found a tight localization of Kon at myotube tips at 18 hr APF, consistent with an important function in attachment initiation (Figure 5A). We also detected β-Int at the myotube tips, but Tsp was absent from myotendinous junctions at 18 hr APF (Figures 5A and S3A), suggesting that integrins are not yet activated at this stage. Consistently, we observed only little Talin and pFAK at myotube tips at 18 hr APF (Figure S3B, C), indicating a cell-cell based contact between muscle and tendon, which was further supported by the localization of E-cadherin to myotube-tendon contacts at 18 hr APF (Figure S3G). During attachment maturation at 30 hr APF, however, Kon and β-Int, as well as Tsp, Talin, and pFAK, strongly accumulated at the myotendinous interface, forming a force-resistant cell-matrix junction (Figures 5D and S3D–S3F).

To investigate the molecular interplay between Kon and integrins, we compared the surface ratios of Kon and β-Int at myotube tips versus the middle region of the myotube. Interestingly, localization of β-Int to myotube tips was less pronounced than that of Kon at 18 hr APF (quantified in Figure S3H). Moreover, β-Int, Talin, and pFAK tip localization was largely lost upon *kon* knockdown (Figures 5B, 5C, and 5E; Figures S3I–S3M), suggesting that Kon is important for integrin accumulation at the myotube tips. The patchy attachments of hypomorphic *kon* knockdown myotubes displayed β-Int concentrations at these patchy myotube-tendon connections (Figure 5F). These data show that Kon contributes to myotube-tendon attachment and suggest that the initial attachment is mediated primarily by direct cell-cell contacts. During attachment maturation, large amounts of ECM and active integrins are recruited, correlating with a stark increase of tension within the myotendinous system and the onset of myofibrillogenesis.

### Attachment Is Essential for Ordered Myofibrillogenesis

As *kon* is required for efficient attachment, we used *kon* knockdown pupae to test our hypothesis that muscle attachment, and thus mechanical tension, influences myofibrillogenesis. Consistent with our live imaging data, we observed readily formed myofibrils spanning the entire length and width of the 150 μm long myofiber at 30 hr APF in wild-type (Figures 6A–6C). Each individual myofibril is connected to both myotendinous junctions at the myofiber ends and contains clusters of the M line protein Obscurin [43], which are regularly spaced at distances of about 1.8 μm along the myofibril (Figures 6B, 6D, and S4Q). Similarly, α-actinin and the titin-like protein kettin, both of which are found at the Z disc in mature muscles [44, 45], show periodic patterns from 30 hr APF onward, suggesting that these myofibrils already house primitive sarcomeres (Figures S4A, S4B, S4G, and S4H). To quantify myofibril length and regularity, we manually traced individual myofibrils in 34 μm × 34 μm × 2.5 μm volumes. We found that both at the fiber surface and in more interior fiber regions, myofibrils spanned regularly throughout the traced segment, suggesting that fibrils reach from one fiber end to the other (Figures 6C and 6M). In contrast, rounded *kon* knockdown muscles displayed severe defects in myofibril assembly. Myofibrils only assembled at the fiber surface, and assembly was abnormal, resulting in fibrils that were too short (Figures 6E–6H and 6M). At a depth of 2.5 μm below the fiber surface, only very short actin aggregates lacking any regular Obscurin pattern could be detected (Figures 6E′–6H′ and 6M). Similarly, α-actinin and kettin patterns were less pronounced in *kon* knockdown IFMs (Figures S4C–S4L). This indicates that attachment or tension, which is gained during attachment maturation, is required for simultaneous myofibrillogenesis throughout the muscle fiber.

To substantiate that a lack of attachment and mechanical tension, and not the rounded muscle shape per se or a possible signal from tendons, indeed causes the observed myofibrillogenesis defects, we investigated myofibrillogenesis in patchily attached *kon* hypomorphic muscles. These fibers showed a defect in fiber compaction at 30 hr APF, leading to 210 μm fibers, compared to 150 μm fibers in wild-type (Figures S2P–S2R). The compaction defect, together with the small number of tendon contacts and the little integrin at the junction (Figure 5F), strongly indicates that less tension is generated within these patchily attached muscles. Importantly, we detected a severe myofibrillogenesis defect in these muscles, with fibrils assembling only close to the fiber surface and not within the fiber interior (Figures 6I–6M). The irregular Obscurin pattern precluded sarcomere length measurements at 30 hr APF in the *kon* hypomorph (Figures 6L and S4Q). Eventually, regular myofibrils and sarcomeres did form in these fibers, possibly because their fiber attachments ultimately matured, leading to thinner but attached fibers at 90 hr APF (Figure 4D). Strikingly, the myofibrils in these fibers were fewer in number and contained very short sarcomeres only 1.7 μm instead of 3.3 μm long. Thus, these sarcomeres resemble "early" sarcomeres of 30 hr APF wild-type fibers that in wild-type gradually grow in length to reach 2.0 μm at 48 hr and 3.3 μm at 90 hr APF (Figures S4M–S4Q). Taken together, these data strongly suggest that Kon and integrins are essential to form mature muscle-tendon attachments, which in turn allow the generation of mechanical tension within the myotendinous system. This tension is essential to initiate myofibrillogenesis simultaneously throughout the myofiber, leading to the highly regular array of myofibrils and sarcomeres of fibrillar IFMs.

### Tension Is Essential for Ordered Myofibrillogenesis

To substantiate that tension buildup rather than an unknown signaling pathway activated by Kon-dependent attachment initiation indeed triggers myofibrillogenesis, we developed an alternative method of tension release from the myotendinous junction after attachment initiation. We applied laser micromanipulations to completely severe the tendon extensions at the anterior end of the myotubes in one thorax half of living pupae after attachment initiation at 22 hr APF. Myofibrillogenesis of the severed thorax half was investigated at 30 hr APF and compared to the intact half, which served as control. We found that the control halves formed regular myofibrils that spanned the entire traced area, suggesting that the myofibrils run from one muscle end to the other (Figures 7A–7D and 7I). In stark contrast, severing of the tendons resulted in defective myofibrillogenesis. Shorter fibrils were detected close to the fiber surface, and only very short actin filaments were present 2.5 μm below the fiber surface (Figures 7E–7I). These data strongly support the *kon* knockdown results, demonstrating that attachment maturation and tension buildup are indeed required to trigger regular simultaneous myofibrillogenesis.

In order to investigate how tension is generated during myofibril assembly and how this tension may trigger myofibrillogenesis, we analyzed flight muscles that express only a headless muscle myosin heavy chain. This mutant Mhc can still assemble into myosin rods but is unable to generate any force [46]. Strikingly, these muscles displayed a robust fiber compaction defect at 30 hr APF (Figures S5A–S5C), strongly suggesting that Mhc contributes to the tension required for myofiber compaction. As a consequence, myofibrillogenesis was strongly abnormal (Figures S5D and S5E). Thus, the regular assembly of bipolar Mhc filaments with crosslinked actin filaments into myofibrils and sarcomeres does contribute to the observed tension within the myotendinous system.

In summary, if tension buildup is blocked by (1) genetically interfering with attachment, (2) severing tendon extensions to muscles, or (3) preventing Mhc motor activity, ordered myofibrillogenesis is strongly defective, demonstrating the requirement for attachment and tension buildup for myofibrillogenesis.

## Discussion

In this paper, we define the temporal hierarchy of *Drosophila* flight muscle-tendon development and establish a causative

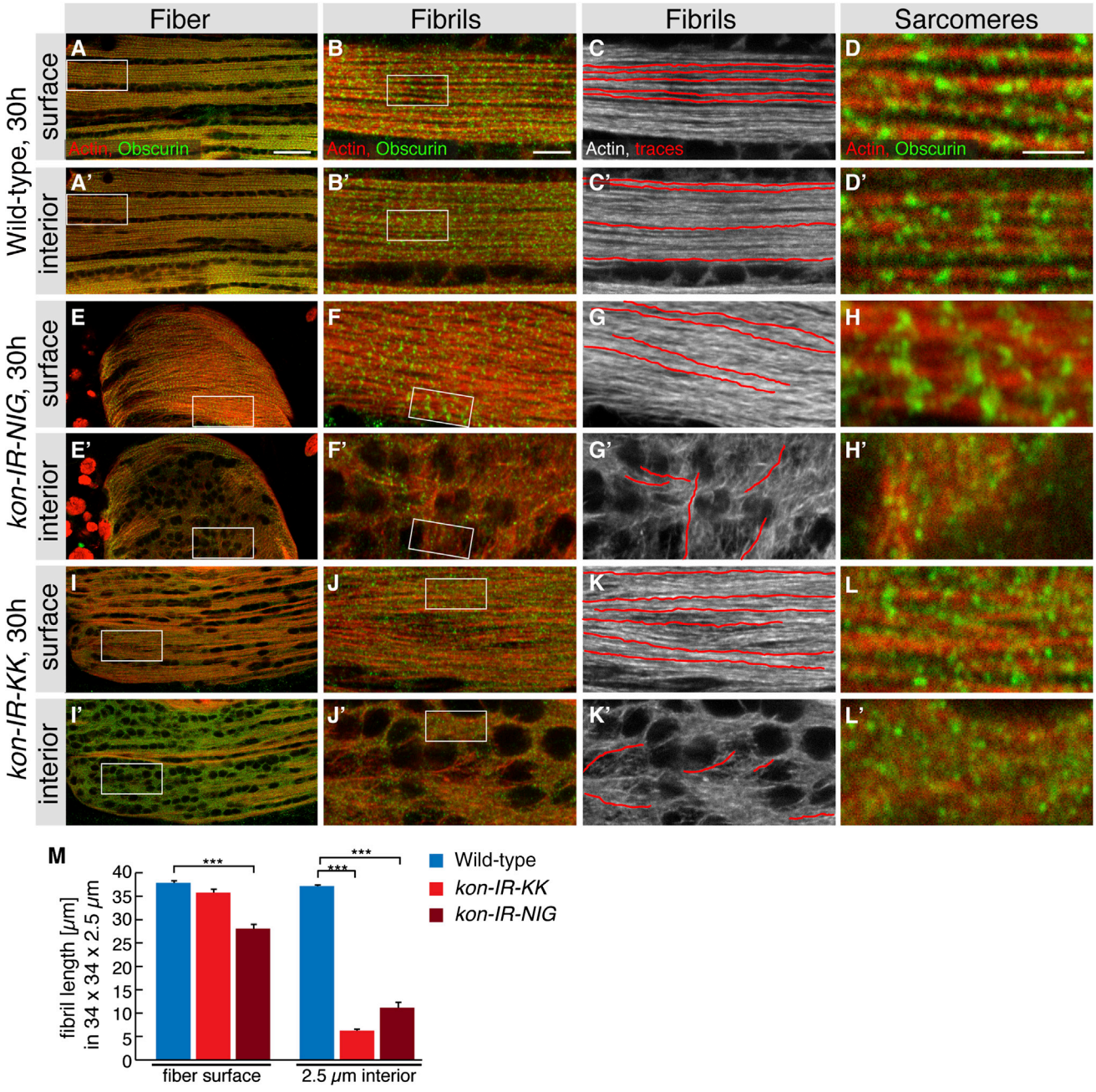



Figure 6. *kon*-Dependent Attachment Is Required for Simultaneous Myofibrillogenesis

(A–L) Surface (A–L) and interior (A′–L′) of DLM myofibers of wild-type (A–D′), *Mef2-GAL4,UAS-kon-IR-NIG* (E–H′), and *UAS-kon-IR-KK;Mef2-GAL4* pupae (I–L′) at 30 hr APF, stained with phalloidin (red) and anti-Obscurin (green). Boxes in (A), (E), and (I) show magnified areas in (B) and (C), (F) and (G), and (J) and (K), respectively; boxes in (B), (F), and (J) are magnified in (D), (H), and (L). Note the regular myofibrils in wild-type in all parts of the fiber, whereas only irregular actin filaments form within the interior of *kon* knockdown fibers. Tracing of fibrils is shown in red in (C), (G), and (K).

(M) Quantification of myofibril length in wild-type and *kon* knockdown pupae at 30 hr APF. Error bars represent SEM, ***$p < 0.001$ (two-tailed unpaired t test). Scale bars represent 20 μm in (A), (E), and (I); 5 μm in (B), (C), (F), (G), (J), and (K); and 2 μm in (D), (H), and (L).

link between muscle-tendon attachment, tension buildup, and myofibrillogenesis. We find that Kon is required for attachment of myotubes to tendon cells. During attachment initiation, few ECM and integrin components are present at the cell-cell contacts of myotubes and tendons, and mechanical tension in the system is low. This is similar to the initial muscle-tendon contacts in the embryo, which are also cell-cell based and do not depend on integrins, and in the case of the ventral-longitudinal muscles also require *kon* [6, 12, 47].

After attachment initiation, more integrin and ECM are recruited to the myotendinous junction, concomitantly with a buildup of mechanical tension in the system. This is in accordance with cell culture data showing that tension on nascent focal adhesions results in integrin clustering and stable integrin-ECM binding, leading to formation of mature focal adhesions [48]. Similarly, increased muscle contractions result in higher integrin-ECM adhesive stability at *Drosophila* larval myotendinous junctions [49]. Thus, it is highly likely that

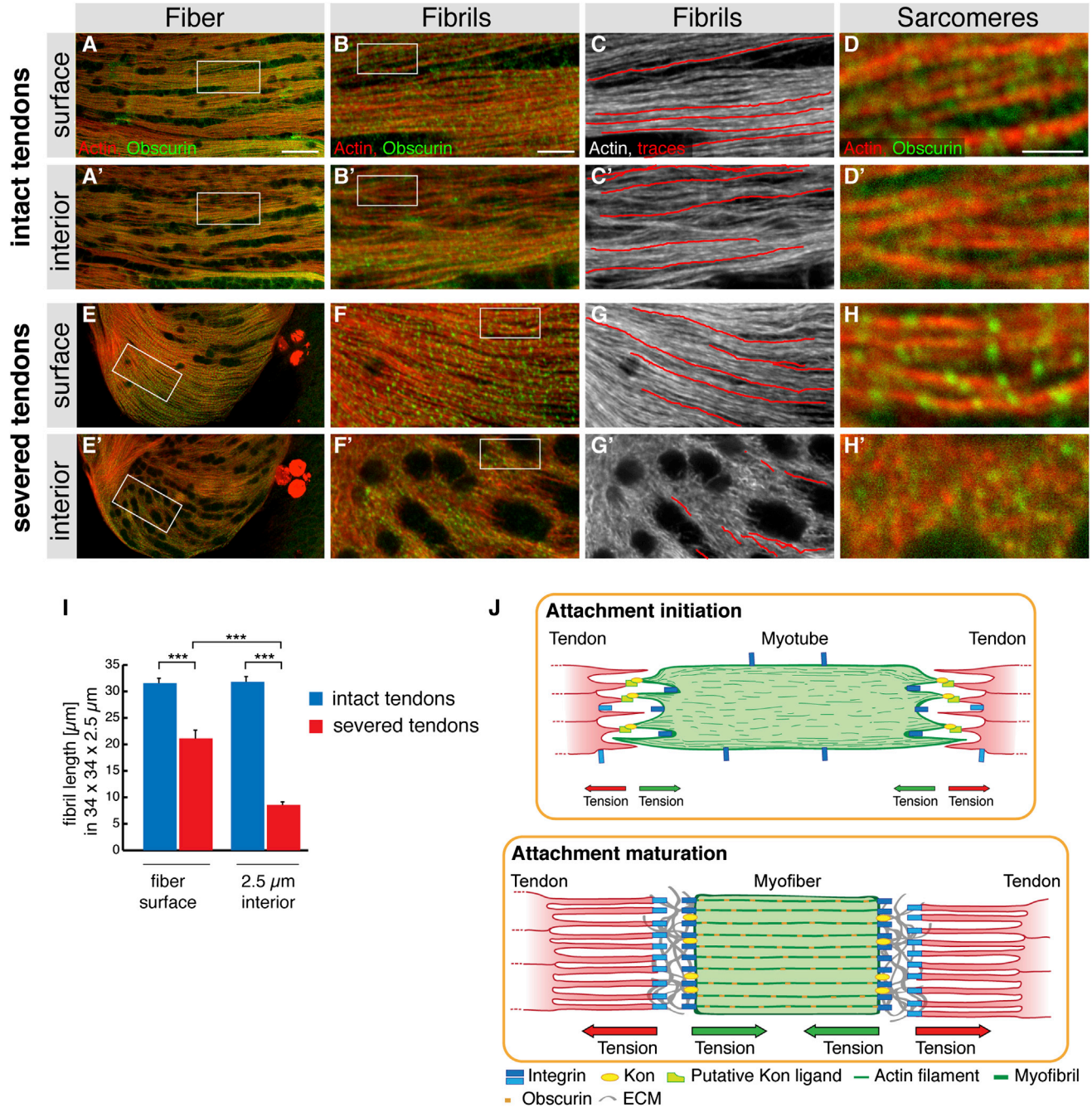



Figure 7. Tension Is Essential for Simultaneous Myofibrillogenesis

(A–H) *UAS-CD8-GFP;Mef2-GAL4,stripe-GAL4* pupae at 30 hr APF with intact (A–D) or laser-severed tendons (E–H). Myofiber surface (A–D and E–H) and interior fiber planes (A′–D′ and E′–H′) are labeled with phalloidin (red) and anti-Obscurin (green). Boxes in (A) and (E) show magnified areas in (B) and (C) and (F) and (G), respectively; boxes in (B) and (F) are magnified in (D) and (H). Myofibril traces are shown in red in (C) and (G). Note that only short, irregular actin filaments are formed within the fiber interior after tendon severing. Scale bars represent 20 μm in (A) and (E); 5 μm in (B), (C), (F), and (G); and 2 μm in (D) and (H).

(I) Quantification of myofibril length in fibers with intact or severed tendons at the fiber surface or interior. Error bars represent SEM, ***$p < 0.001$ (two-tailed unpaired t test).

(J) Schematic model of attachment initiation and attachment maturation showing tension buildup and regularly arrayed myofibrils.

integrins cluster and stably anchor the flight muscle fibers to the ECM when tension is built up. A similar role for integrins at *Drosophila* larval and vertebrate myotendinous junctions has been suggested; however, mechanical tension has not been measured directly in these systems [10].

Integrins link to the actin cytoskeleton via intracellular adaptors [50]. Thus, activated integrins are likely connecting the assembling myofibrils to the ECM and transmitting tension to the tendon extensions, resulting in their dramatic elongation. An increase in tension will in turn recruit more integrins,

resulting in a more stable link to the assembling myofibrils. If the muscle is rounded after efficient *kon* knockdown or tendon severing, or is not homogenously attached at both ends, as in *kon* hypomorphic muscles, integrin accumulation is defective. Integrins are only localized to the few patchy muscle-tendon contacts and possibly to the surface of the rounded fibers by interaction with surrounding ECM. In both cases, this should lead to a very inhomogeneous force field that only enables local myofibril formation close to the patchy attachments or the fiber surface, instead of simultaneous myofibrillogenesis throughout the entire fiber. It will be a challenge for future research to utilize force sensors in integrin complex components to directly measure molecular forces during myofibrillogenesis in vivo. Force measurements on such components at focal adhesions in cell culture offer interesting future perspectives [27].

In addition to controlling local assembly of protein complexes, tension is also an ideal way to coordinate developmental decisions over long distances in large cells or cell networks. An actomyosin array, for example, synchronizes dorsal closure at the leading edge of the *Drosophila* epidermis [51, 52] and maintains a stable compartment boundary in wing imaginal discs [53]. The advantage of tension control is particularly obvious when the goal of the process is the simultaneous assembly of a large cytoskeletal ensemble—the assembly of many thousand sarcomeres during myofibrillogenesis in a 150 μm muscle fiber.

Many models of myofibrillogenesis rely on differentiating myoblasts in cell culture [13, 23, 36]. However, it is difficult to address the role of tension in myofibrillogenesis using these in vitro models, as the differentiating cells adhere more or less homogenously to the dish instead of selectively to tendon cells at both muscle fiber ends. Thus, tension is dispersed in all directions in vitro, whereas it is oriented along the long axis of the myofiber in vivo. Our revised model of myofibrillogenesis proposes that tension triggers myofibrillar self-organization. This model requires muscle-tendon attachment via integrins in order to build up directed tension and thus coordinates attachment with myofibrillogenesis (Figure 7J). The evidence supporting our model is 5-fold:

(1) Integrin recruitment to the junction coincides with a stark increase in tension during attachment maturation that precedes myofibrillogenesis.
(2) The simultaneously assembled myofibrils always connect from one integrin-containing junction to the other throughout the entire myofiber.
(3) An inhomogeneous force field generated by patchy attachment does not allow proper myofibrillogenesis. Abnormal myofibrils only form close to the patchy attachments.
(4) Releasing tension by efficient *kon* knockdown or laser-induced severing of the tendons leads to major myofibrillogenesis defects.
(5) Simultaneous myofibril assembly requires Mhc motor activity that itself contributes to tension formation.

The observed simultaneous myofibril assembly strongly supports the hypothesis that several multiprotein complexes [24] or possibly I-Z-I and thick filament precursors [22] self-organize into distinct myofibrils. Theoretical predictions that suggest that bipolar actin and myosin filaments require a certain amount of directed tension to pass the threshold of self-assembly initiation [5] are also consistent with our model.

Similar to *Drosophila*, vertebrate skeletal muscles also require the formation of stable attachments before they assemble myofibrils, and cardiomyocytes grown in suspension cannot assemble a mature myofibrillar apparatus [54]. Interestingly, myofibril assembly also appears to depend on integrin function in skeletal muscle [20]. Thus, we speculate that a similar tension-based mechanism also functions in vertebrate muscles to assemble the core, evolutionarily conserved sarcomeric components into regular periodic myofibrils.

## Supplemental Information

Supplemental Information includes five figures, Supplemental Experimental Procedures, and seven movies and can be found with this article online at http://dx.doi.org/10.1016/j.cub.2014.02.032.

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